MA2040 Jan-Apr 2024
Problem Sheet 6

1. Suppose $X_1, \ldots, X_n$ are i.i.d. Poisson random variables with mean λ. Estimate λ using both method of moments and maximum likelihood estimator.

2. Suppose $X_1, \ldots, X_n$ are i.i.d. normal with mean μ and variance σ^2. Determine maximum likelihood estimator for σ^2 when μ is known.

3. Suppose $X_1, \ldots, X_n$ are i.i.d. normal with mean μ and variance σ^2. Find the upper and lower confidence intervals for mean when variance is known.

4. Suppose $X_1, \ldots, X_n$ are i.i.d. normal with mean μ and variance σ^2. Find the upper and lower confidence intervals for mean when variance is unknown.

5. Suppose $X_1, \ldots, X_n$ are i.i.d. normal with mean μ and variance σ^2. Let $V = \frac{1}{n}\sum(X_1 - \bar{X})^2$ be an estimate for σ^2. Find the bias and m.s.e. for V.

6. Suppose $X_1, \ldots, X_n$ are i.i.d. Bernoulli with mean p. Find the bias and m.s.e. for the estimator $\bar{X}$.